

HR Employee Database System

Database Design & Implementation Project

Microsoft SQL Server (T-SQL) Edition

2026

1. Project Overview

The HR Employee Database System is a relational database solution designed to centralize and manage all essential Human Resources information. This document covers the full implementation using Microsoft SQL Server (T-SQL), including table design, constraints, computed columns, stored procedures, and reporting queries. The system handles employee records, attendance tracking, leave management, payroll processing, training programs, and role-based access control.

1.1 Objectives

- Centralize and store complete employee data securely.
- Automate key HR functions: attendance, leave management, and payroll.
- Provide fast and accurate reports for HR managers and leadership.
- Ensure data integrity, confidentiality, and controlled access.
- Improve HR operational efficiency by reducing manual processes.
- Support future scalability and integration with other systems.

1.2 Scope

The system covers the following modules:

- Employee Information Management
- Attendance & Leave Tracking
- Payroll Management
- Training & Development
- Reporting Module
- Role-Based Access Control

2. Entity Relationship Diagram (ERD)

2.1 Entities and Relationships

The following describes all entities, their attributes, and the relationships between them. The ERD can be visualized in any tool such as SQL Server Management Studio (SSMS) Database Diagram, draw.io, or Lucidchart using the structure below.

2.2 Entity List

- **Departments**
- **Employees**
- **Payroll**
- **Attendance**
- **LeaveTypes**
- **LeaveRequests**
- **LeaveBalance**
- **TrainingPrograms**
- **TrainingEnrollment**
- **Users**

2.3 Relationships

Entity A	Entity B	Relationship
Departments	Employees	One-to-Many: A department has many employees
Employees	Departments	Many-to-One: Manager (FK) is an Employee
Employees	Payroll	One-to-Many: An employee has many payroll records
Employees	Attendance	One-to-Many: An employee has many attendance logs
Employees	LeaveRequests	One-to-Many: An employee submits many leave requests
LeaveTypes	LeaveRequests	One-to-Many: A leave type categorizes many requests
Employees	LeaveBalance	One-to-Many: An employee has balance per leave type
LeaveTypes	LeaveBalance	One-to-Many: Each leave type has balances per employee
Employees	TrainingEnrollment	Many-to-Many (via TrainingEnrollment)
TrainingPrograms	TrainingEnrollment	Many-to-Many (via TrainingEnrollment)
Employees	Users	One-to-One: Each employee may have one system user

Key Cardinalities:

- Departments to Employees: 1 to Many (One department, many employees)
- Employees to Payroll: 1 to Many (Monthly payroll records)
- Employees to LeaveRequests: 1 to Many
- Employees to TrainingPrograms: Many to Many (through TrainingEnrollment)
- Employees to Users: 1 to 1 (One login per employee)

3. Database Dictionary

This section describes every table, its columns, T-SQL data types, constraints, and purpose. All string columns use NVARCHAR (supports Unicode and Arabic text). Auto-increment uses IDENTITY(1,1). Computed columns use AS (...) PERSISTED. ENUM is replaced by NVARCHAR + CHECK constraints. BOOLEAN is replaced by BIT.

Table: Departments

Column Name	Data Type	PK	FK	Description / Notes
DeptID	INT	PK		Auto-increment primary key
DeptName	NVARCHAR(100)			Name of the department (NOT NULL)
ManagerID	INT		FK→Employees	Employee who manages this department
Location	NVARCHAR(100)			Physical location of the department
CreatedAt	DATETIME2			Timestamp when record was created

Table: Employees

Column Name	Data Type	PK	FK	Description / Notes
EmpID	INT	PK		Auto-increment primary key
FirstName	NVARCHAR(50)			Employee first name (NOT NULL)
LastName	NVARCHAR(50)			Employee last name (NOT NULL)
NationalID	NVARCHAR(20)			National ID number (UNIQUE, NOT NULL)
BirthDate	DATE			Date of birth
Gender	NVARCHAR(10) CHECK			Male / Female / Other
Email	NVARCHAR(100)			Work email address (UNIQUE)
Phone	NVARCHAR(20)			Contact phone number
Address	NVARCHAR(255)			Residential address
HireDate	DATE			Date of employment (NOT NULL)
JobTitle	NVARCHAR(100)			Current job title
DeptID	INT		FK→Departments	Department the employee belongs to
GradeLevel	NVARCHAR(10)			Salary/grade classification (G1–G7)
EmploymentType	NVARCHAR(20) CHECK			Full-Time / Part-Time / Contract
Status	NVARCHAR(20) CHECK			Active / Inactive / Terminated

Table: Payroll

Column Name	Data Type	PK	FK	Description / Notes
PayrollID	INT	PK		Auto-increment primary key
EmpID	INT		FK→Employees	Employee reference
PayPeriod	DATE			First day of the pay month
BasicSalary	DECIMAL(10,2)			Base monthly salary (NOT NULL)
HousingAllow	DECIMAL(10,2)			Housing allowance
TransportAllow	DECIMAL(10,2)			Transportation allowance
OtherAllow	DECIMAL(10,2)			Other allowances
TaxDeduction	DECIMAL(10,2)			Income tax deduction
InsuranceDed	DECIMAL(10,2)			Social insurance deduction
OtherDeduct	DECIMAL(10,2)			Other deductions
NetSalary	DECIMAL(10,2) PERSISTED			PERSISTED computed: BasicSalary + Allowances - Deductions
ProcessedDate	DATETIME2			Timestamp when payroll was processed

Table: Attendance

Column Name	Data Type	PK	FK	Description / Notes
AttendID	INT	PK		Auto-increment primary key
EmpID	INT		FK→Employees	Employee reference
AttDate	DATE			Date of attendance record
CheckIn	TIME			Time employee checked in
CheckOut	TIME			Time employee checked out
Status	NVARCHAR(20) CHECK			Present / Absent / Late / Half-Day / Holiday
Notes	NVARCHAR(255)			Optional notes or reason

Note: (EmpID, AttDate) has a UNIQUE constraint to prevent duplicate daily records. T-SQL MERGE statement is used in sp_LogAttendance for UPSERT behavior (replaces MySQL ON DUPLICATE KEY UPDATE).

Table: LeaveTypes

Column Name	Data Type	PK	FK	Description / Notes
LeaveTypeID	INT	PK		Auto-increment primary key
TypeName	NVARCHAR(50)			Name of leave type (e.g., Annual, Sick)
MaxDaysPerYear	INT			Maximum allowed days per year
IsPaid	BIT			Whether this leave type is paid

Table: LeaveRequests

Column Name	Data Type	PK	FK	Description / Notes
LeaveID	INT	PK		Auto-increment primary key
EmpID	INT		FK→Employees	Employee who submitted the request
LeaveTypeID	INT		FK→LeaveTypes	Type of leave requested
StartDate	DATE			Start date of leave
EndDate	DATE			End date of leave
TotalDays	INT PERSISTED			PERSISTED computed: DATEDIFF(DAY, StartDate, EndDate)+1
Reason	NVARCHAR(500)			Employee-provided reason
Status	NVARCHAR(20) CHECK			Pending / Approved / Rejected
ApprovedBy	INT		FK→Employees	Manager who approved/rejected
RequestDate	DATETIME2			When the request was submitted

Table: LeaveBalance

Column Name	Data Type	PK	FK	Description / Notes
BalanceID	INT	PK		Auto-increment primary key
EmpID	INT		FK→Employees	Employee reference
LeaveTypeID	INT		FK→LeaveTypes	Leave type reference
Year	INT			Calendar year
TotalAllocated	INT			Days allocated for the year
TotalUsed	INT			Days actually taken (approved)
Remaining	INT PERSISTED			PERSISTED computed: TotalAllocated - TotalUsed

Table: TrainingPrograms

Column Name	Data Type	PK	FK	Description / Notes
ProgramID	INT	PK		Auto-increment primary key
ProgramName	NVARCHAR(150)			Name of training program (NOT NULL)
Description	NVARCHAR(MAX)			Detailed description of the program
StartDate	DATE			Program start date
EndDate	DATE			Program end date
Location	NVARCHAR(100)			Training venue or 'Online'
Trainer	NVARCHAR(100)			Name of trainer or organization
MaxCapacity	INT			Maximum number of participants

Table: TrainingEnrollment

Column Name	Data Type	PK	FK	Description / Notes
EnrollID	INT	PK		Auto-increment primary key
EmpID	INT		FK→Employees	Employee who enrolled
ProgramID	INT		FK→TrainingPrograms	Training program reference
EnrollDate	DATE			Date of enrollment
Status	NVARCHAR(20) CHECK			Enrolled / Completed / Cancelled
Score	DECIMAL(5,2)			Final assessment score (nullable)
Certificate	NVARCHAR(255)			Certificate number or file path

Table: Users

Column Name	Data Type	PK	FK	Description / Notes
UserID	INT	PK		Auto-increment primary key
EmpID	INT		FK→Employees	One-to-one with Employees (UNIQUE)
Username	NVARCHAR(50)			Unique login username
PasswordHash	NVARCHAR(255)			Hashed password (bcrypt)
Role	NVARCHAR(20) CHECK			Admin / HR / Manager / Employee
IsActive	BIT			Whether user account is active
LastLogin	DATETIME2			Timestamp of last successful login

4. Stored Procedures

All data manipulation is performed through stored procedures to ensure consistency, security, and maintainability.

4.1 Employee Management

Procedure	Operation	Description
sp_GetAllEmployees()	SELECT	Returns all active employees with department info
sp_GetEmployeeByID(EmpID)	SELECT	Returns full details for a specific employee
sp_InsertEmployee(...)	INSERT	Adds a new employee record — returns new ID via SCOPE_IDENTITY()
sp_UpdateEmployee(...)	UPDATE	Updates job details, contact, and status
sp_DeleteEmployee(EmpID)	UPDATE	Soft-deletes: marks employee as Terminated
sp_SearchEmployees(Keyword)	SELECT	Full-text search across name, email, job, department

4.2 Attendance Management

Procedure	Operation	Description
sp_LogAttendance(...)	INSERT/UPDATE	Logs or updates daily attendance using T-SQL MERGE (UPSERT)
sp_RecordCheckOut(...)	UPDATE	Records employee check-out time
sp_GetAttendanceByMonth(...)	SELECT	Retrieves attendance records for a month

4.3 Leave Management

Procedure	Operation	Description
sp_SubmitLeaveRequest(...)	INSERT	Validates balance, then submits leave request
sp_ProcessLeaveRequest(...)	UPDATE	Approves/rejects request and updates balance
sp_GetLeaveBalance(EmpID,Year)	SELECT	Shows leave balance by type for an employee

4.4 Payroll Management

Procedure	Operation	Description
sp_InsertPayroll(...)	INSERT	Records monthly payroll with all components
sp_GetPayHistory(EmpID)	SELECT	Returns full pay history for an employee

4.5 Training Management

Procedure	Operation	Description
sp_EnrollTraining(EmpID,ProgramID)	INSERT	Enrolls employee, checks capacity first
sp_CompleteTraining(...)	UPDATE	Records completion with score and certificate

4.6 Report Procedures

Report Procedure	Output Description
rpt_EmployeeMasterList()	Full employee list with department, grade, years of service
rpt_AttendanceSummary(Year,Month)	Monthly attendance counts per employee (Present/Absent/Late/Half-Day)
rpt_PayrollSummary(PayPeriod)	Monthly payroll with allowances, deductions, and net salary
rpt_LeaveBalanceReport(Year)	Leave balance per employee per leave type
rpt_TrainingReport()	Training enrollment, completion status, and scores

5. How to Run the Database

5.1 Requirements

- Microsoft SQL Server 2019 or 2022 (Express edition is free)
- SQL Server Management Studio — SSMS (free download from Microsoft)
- The file: HR_Database_MSSQL.sql

5.2 Setup Steps

1. Open SQL Server Management Studio (SSMS) and connect to your SQL Server instance.
2. Go to File > Open > File and select HR_Database_MSSQL.sql.
3. Press F5 or click the Execute button to run the entire script.
4. The script will: Create the HR_System database, create all 10 tables with constraints, insert sample data for 8 employees across 5 departments, and create all stored procedures.
5. To test, scroll to Section 6 in the SQL file and execute the example EXEC statements.

5.3 Sample Procedure Calls

```
-- View all employees EXEC sp_GetAllEmployees;
-- Search by keyword EXEC sp_SearchEmployees @Keyword = N'Ahmed';
-- Get January 2024 attendance for employee 1 EXEC sp_GetAttendanceByMonth @EmpID = 1, @Year
= 2024, @Month = 1;
-- Submit a leave request EXEC sp_SubmitLeaveRequest @EmpID = 4, @LeaveTypeID = 1, @StartDate
= '2024-07-01', @EndDate = '2024-07-05', @Reason = N'Vacation';
-- Approve leave request ID 5 by manager (EmpID 3) EXEC sp_ProcessLeaveRequest @LeaveID = 5,
@ApproverID = 3, @Decision = N'Approved';
-- Generate payroll report for January 2024 EXEC rpt_PayrollSummary @PayPeriod = '2024-01-
01';
-- Generate employee master list EXEC rpt_EmployeeMasterList;
```

6. Security & Access Control

6.1 Role-Based Access

The Users table implements role-based access control with four roles:

Role	Permissions
Admin	Full access to all tables and procedures including user management
HR	Access to employee records, attendance, leave management, and all reports
Manager	Can approve/reject leave requests, view team attendance and reports
Employee	Can view own records, submit leave requests, and enroll in training

6.2 Security Practices

- Passwords are stored as bcrypt hashes (never plain text).
- Soft delete is used for employees (Status = Terminated) to preserve data integrity.
- Foreign key constraints ensure referential integrity across all tables.
- PERSISTED computed columns (NetSalary, TotalDays, Remaining) prevent manual calculation errors.
- UNIQUE constraints prevent duplicate attendance records and user accounts.

7. Project Summary

Item	Details
Total Tables	10 tables
Total Stored Procedures	17 procedures (12 DML + 5 Reports)
Sample Employees	8 employees across 5 departments
Database Engine	Microsoft SQL Server 2019 / 2022
Key Features	PERSISTED computed columns, MERGE (UPSERT), CHECK constraints, IDENTITY, SCOPE_IDENTITY(), soft delete
Reports Available	Employee Master List, Attendance Summary, Payroll, Leave Balance, Training

This system provides a solid foundation for an HR department, covering the most critical operational needs with room to expand into performance management, recruitment, and payroll automation in future phases.

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